

Project Idea: CrowdFlow Studio

1. Project Idea: What We Are Trying to Do

CrowdFlow Studio is a professional, browser-based crowd-simulation and venue-modeling platform. We are building an end-to-end digital twin ecosystem that allows users to import, design, edit, and simulate human movement within any spatial environment—from intimate corporate event halls to stadiums, airports, and festival grounds.

The core objective is to deliver a real-world, industry-ready application that:

- Automatically reconstructs spatial environments from 2D images or CAD files using AI.
- Provides a manual, web-based Canvas Editor for custom venue building.
- Facilitates customized path and behavior modeling using intelligent simulation components.
- Executes accurate, agent-based microscopic crowd simulations directly in the browser.
- Synthesizes resulting data into actionable analytical outputs (heatmaps, flow-lines) and professional safety reports.

2. Problem Statement: What We Are Trying to Solve

The global market for visitor flow simulation is projected to surge from \$1.2 billion in 2024 to \$4.8 billion by 2033.¹ Despite this, the software ecosystem remains highly fragmented.

We are solving a pronounced dichotomy in the industry:

- **The Over-Engineered High End:** Legacy micro-simulation platforms (e.g., Pathfinder, LEGION) deliver highly accurate agent-based modeling but require specialized engineering expertise, heavy local hardware, and prolonged setup phases.¹ They are inaccessible to standard event coordinators or mid-market safety officers.
- **The Under-Powered Low End:** Event-focused layout platforms (e.g., OnePlan, Iventis) offer intuitive, drag-and-drop browser interfaces but lack dynamic simulation engines.¹ They rely on static mathematical equations for capacity, completely omitting the fluid dynamics of human movement, bottleneck formation, and panic behavior.¹

CrowdFlow Studio solves this by bridging the gap: marrying intuitive, automated venue design with a professional-grade, high-performance simulation engine accessible entirely via a standard web browser.

3. Literature Review

The development of CrowdFlow Studio intersects two primary academic and industrial domains: pedestrian dynamics modeling and automated architectural vectorization.

Crowd Simulation and Pedestrian Dynamics: The foundational algorithm for microscopic pedestrian behavior is the Social Force Model (SFM), introduced by Helbing and Molnar in 1995, which utilizes coupled Langevin equations to simulate individuals reacting to attractive goals and repulsive obstacles. While traditional SFM accurately captures lane formation and congestion, scaling it to massive crowds has historically required heavy desktop CPU processing. Recent advancements have explored Position-Based Dynamics and GPU parallelization to handle dense crowds interactively. However, compiling these physics models directly into WebAssembly (WASM) via Rust to bypass traditional JavaScript or Unity engine overhead has emerged as a cutting-edge technique capable of simulating over 100,000 agents in real-time within a web environment.

Automated Floor Plan Vectorization: Historically, generating simulation-ready environments from 2D floor plans required manual CAD tracing. Recent deep learning approaches have automated this. Methods now utilize frameworks combining YOLO for axis grid detection with graph-based topological reconstruction to generate editable CAD lines. To handle complex architectural variations, researchers utilize datasets like CubiCasa5k alongside hybrid architectures (such as SegFormer combined with CNN-based U-Net decoders) to accurately extract fine-grained wall segments and structural boundaries. More advanced generative pipelines, such as Raster2Seq, treat floor plan reconstruction as an image-conditioned sequence modeling task, autoregressively outputting labeled polygon sequences to mimic natural CAD design workflows.

4. Technical Details & Final Product Architecture (Brief)

To make the final product seamless for the end-user, CrowdFlow Studio relies on a highly advanced, split-architecture technical stack:

- **AI Venue Import System (Cloud Backend):** Users upload DWG, PDF, or raster images. The backend utilizes Vision-Language Models (VLMs) and sequence modeling to execute raster-to-vector reconstruction. Automated geometric cleaning algorithms snap walls, close topological gaps, and purge scanning artifacts to generate a mathematically sound navigation mesh.
- **Canvas Editor & Component Library (React Frontend):** A responsive, browser-based 2D editor where the imported vector layout becomes manipulable. Users can drag and drop intelligent simulation elements (turnstiles, security checkpoints, stairs) that are encoded with physical dimensions and throughput limits (e.g., processing times,

maximum persons-per-minute).

- **High-Performance Simulation Engine (Rust/WebAssembly):** The computational nucleus is built in Rust and compiled to WebAssembly (WASM), bypassing JavaScript's single-threaded limitations. By utilizing an Entity-Component-System (ECS) memory architecture, the engine ensures strict cache locality and maximum CPU utilization. This allows the browser to run complex Social Force Models, collision avoidance, and pathfinding for tens of thousands of agents at high frame rates.

5. Final Product Vision & Real-World Impact

The final product functions as a proactive risk mitigation and commercial optimization dashboard. Following a simulation run, the system automatically translates raw agent coordinates into Density Heatmaps, Flow-Line Paths, and Bottleneck Detection graphs.

Real-World Impact: Beyond logistical optimization, the tool directly addresses strict emerging life-safety compliance requirements. For example, the UK's Terrorism (Protection of Premises) Act 2025 (widely known as Martyn's Law) now legally requires venues and events hosting 800 or more individuals to assess vulnerabilities, implement strict lockdown procedures, and ensure robust evacuation routes are in place. CrowdFlow Studio's automated regulatory engine—which integrates National Fire Protection Association (NFPA 101) algorithms and UK Green Guide egress flow rates—allows safety officers to mathematically validate their evacuation times against these new legal frameworks and export professional compliance reports instantly.

6. Potential Publications and Patent Scopes

Potential Publications: The interdisciplinary nature of this project offers several strong avenues for academic and industry publication:

1. **Computer Graphics & Web Technologies:** A paper detailing the specific integration of Rust-WASM with an Entity-Component-System (ECS) for zero-copy memory handoffs in massive browser-based agent simulations.
2. **Civil Engineering & Automated Modeling:** A methodology paper demonstrating the hybrid VLM-to-Vector pipeline for converting rasterized legacy floor plans into topologically closed simulation meshes (Targeting journals like *MDPI Applied Sciences* or the *Journal of Computing in Civil Engineering*).

Patent Scopes: There is significant intellectual property potential in the proprietary workflows developed for this platform:

1. **Automated Environment Generation:** Patenting the specific pipeline that combines AI raster sequence modeling with algorithmic topological gap-closing specifically for the generation of multi-agent navigation meshes. (This builds upon, but differentiates from, existing patents like EP4242587B1, which covers the broader automation of drawings to

indoor maps).

2. **Browser-Based Simulation Memory Architecture:** Patenting the specific implementation of the Web Worker and SharedArrayBuffer memory alignment that allows real-time rendering of massive crowd density heatmaps entirely client-side without relying on cloud-compute streaming.